

Intel Introduces Thunderbolt 5 Connectivity Standard

The industry-leading connectivity solution will provide up to three times more bandwidth compared with Thunderbolt 4.



News

September 12, 2023

Contact Intel PR

Follow Intel Newsroom on social:



More Client Computing
News

By

What's New: Today, Intel announced Thunderbolt™ 5 – the next generation of Thunderbolt – and demonstrated a prototype laptop and dock. Thunderbolt 5 promises to deliver significant improvements in connectivity speed and bandwidth benefits for computer users.

 author-image



"Thunderbolt 5 will provide industry-leading performance and capability for connecting computers to monitors, docks, storage and more. Intel is excited to continue our tradition of leadership for wired connectivity solutions. Thunderbolt is now the mainstream port for connectivity on mobile PCs, and delivering the next generation of performance with Thunderbolt 5 will provide even more capability for the most demanding users."

–Jason Ziller, general manager of the Client Connectivity Division at Intel

What It Does: Thunderbolt 5 will deliver 80 gigabits per second (Gbps) of bi-directional bandwidth, and with Bandwidth Boost it will provide up to 120 Gbps for the best display experience. These improvements will provide up to three times more bandwidth than the best existing connectivity solution, providing outstanding display and data connections. Thunderbolt 5 will meet the high bandwidth needs of content creators and gamers. Built on industry standards – including USB4 V2 – Thunderbolt 5 will be broadly compatible with previous versions of Thunderbolt and USB.

Why It Matters: Bandwidth needs of content creators, gamers and professionals are increasing significantly. These users want high-resolution displays and low-latency visuals while working with increasingly larger video and data files. Thunderbolt 5 has been designed to massively improve connectivity speed and bandwidth to ensure modern PC users can enjoy the highest-quality visuals and immersive experiences for years to come.

"Microsoft is excited to closely collaborate with Intel to bring the latest USB4 standard to Windows customers," said Ian LeGrow, corporate vice president of Core OS product management at Microsoft Corp. "Thunderbolt 5 is fully USB 80Gbps standard compliant to support the next generation of high-performance displays, storage and connectivity."

New Capabilities Powered by Thunderbolt™ 5



Thunderbolt 5 has been designed to massively improve connectivity speed and bandwidth to ensure modern PC users can enjoy the highest-quality visuals and immersive experiences for years to come. (Credit: Intel Corporation)

Thunderbolt 5 will deliver 80 gigabits per second (Gbps) of bi-directional bandwidth, and with Bandwidth Boost it will provide up to 120 Gbps for the best display experience. These improvements will provide up to three times more bandwidth than the best existing connectivity solution, providing outstanding display and data connections. Thunderbolt 5 will meet the high bandwidth needs of content creators and gamers. (Credit: Intel Corporation)

Thunderbolt 5 builds upon Thunderbolt 4 in several ways, including:

- Two times the total bi-directional bandwidth; Bandwidth Boost provides up to three times the throughput for video-intensive usage, up to 120 Gbps.
- Double the PCI Express data throughput for faster storage and external graphics.
- Built on industry standards including USB4 V2, DisplayPort 2.1 and PCI Express Gen 4; fully compatible with previous versions.
- Double the bandwidth of Thunderbolt Networking for high-speed PC-to-PC connections.
- Utilizes a new signaling technology, PAM-3, to deliver these significant increases in performance with today's printed circuit boards, connectors and passive cables up to 1 meter.

About Thunderbolt in the PC Market:

Thunderbolt products have become mainstream in the PC marketplace, with leading PC accessory vendors worldwide adopting the standard to reach hundreds of millions of PC users active today. Thunderbolt 5 will build on this extensive base and continue to deliver on the vision of one USB-C port

and cable that can do it all – simplifying the PC experience for users around the globe while still providing the highest quality and performance possible.

The announcement of Thunderbolt 5 is another milestone in Intel's long history of leading the industry in input/output (I/O) innovation. Intel works closely with PC, accessory and cable partners to deliver the most advanced and complete wired connectivity solutions through dedicated enabling and testing programs.

All Thunderbolt products undergo stringent certification testing to ensure the best wired connection solution performance for the PC industry. Products that pass this testing use the Thunderbolt brand royalty-free. That's why Thunderbolt has become a globally recognized indicator of the best wired connectivity solutions for PCs and accessories.

What's Next: Computers and accessories based on Intel's Thunderbolt 5 controller, code-named Barlow Ridge, are expected to be available starting in 2024.

More Context: [Intel Thunderbolt Technology](#)

|
[Engineered for Intel® Evo™ Accessory Program](#)

| [Thunderbolt 5:
The Next Generation -- The Best Solution for
Creators and Gamers](#)

(Media Deck)

Tags

Client Computing

About Intel

Intel (Nasdaq: INTC) is an industry leader, creating world-changing technology that enables global progress and enriches lives. Inspired by Moore's Law, we continuously work to advance the design and manufacturing of semiconductors to help address our customers' greatest challenges. By

embedding intelligence in the cloud, network, edge and every kind of computing device, we unleash the potential of data to transform business and society for the better. To learn more about Intel's innovations, go to newsroom.intel.com and intel.com.

© Intel Corporation. Intel, the Intel logo and other Intel marks are trademarks of Intel Corporation or its subsidiaries. Other names and brands may be claimed as the property of others.

[Company Overview](#)

[Contact Intel](#)

[Newsroom](#)

[Investors](#)

[Careers](#)

[Corporate Responsibility](#)

[Diversity & Inclusion](#)

[Public Policy](#)



© Intel Corporation

[Terms of Use](#)

[*Trademarks](#)

[Cookies](#)

[Privacy](#)

[Supply Chain Transparency](#)

[Site Map](#)

[Recycling](#)

Intel technologies may require enabled hardware, software or service activation. // No product or component can be absolutely secure. // Your costs and results may vary. // Performance varies by use, configuration and other factors. // See our complete legal [Notices and Disclaimers](#)

. // Intel is committed to respecting human rights and avoiding causing or contributing to adverse impacts on human rights. See Intel's [Global Human Rights Principles](#). Intel's products and software are intended only to be used in applications that do not cause or contribute to adverse impacts on human rights.

